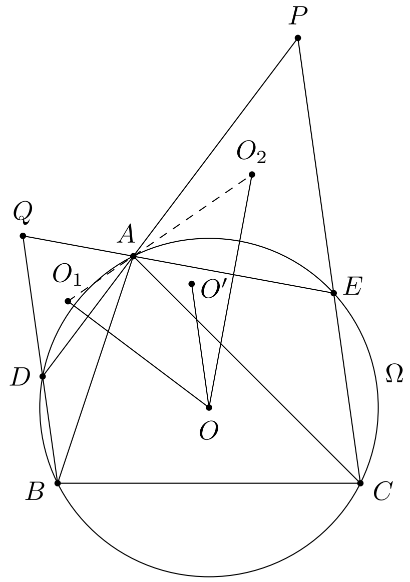
HK IMO Training 2023-2024
Problem Set 13 Suggested Solutions

Problem 49 (Saudi Arabia Training 2019). All the positive integers are coloured in 3 colours randomly. Prove that there are two positive integers of the same colour such that their difference is a perfect square.

Solution. Suppose we cannot find these two positive integers. Then for any positive integer n, the colours of $n, n+9, n+25$ are pairwise distinct since the difference of any two of them is a perfect square. Similarly, the colours of $n, n+16, n+25$ are pairwise distinct. As there are only 3 different colours, $n+9$ and $n+16$ must be of the same colour. Equivalently, for every integer $m \geqslant 10$, m and $m+7$ must have the same colour. Inductively, m has the same colour as $m+7k$ for any positive integer k. In particular, m has the same colour as $m+49$, which is a contradiction. Thus, there must be two positive integers of the same colour such that their difference is a perfect square. $\square$

Problem 50 (European Mathematical Cup 2019 Junior Problem 3). Let Ω be the circumcircle of $\triangle ABC$. Let ℓ_1 and ℓ_2 be parallel straight lines passing through B and C respectively. Suppose ℓ_1 and ℓ_2 meet Ω at D and E respectively, such that both D and E lie on the same side of the line BC as A. The lines AD and CE meet at P. The lines AE and BD meet at Q. Let O, O_1, O_2, O' be the circumcentres of $\triangle ABC$, $\triangle ADQ$, $\triangle AEP$ and $\triangle OO_1O_2$ respectively. Prove that $OO' \mathbin{/\!/} \ell_1$.



Solution. We have

$$\angle O_1AD = 90^\circ - \angle DQA = 90^\circ - \angle PEA = \angle O_2AP.$$

This shows O_1, A, O_2 are collinear. Next, as OO_2 is the perpendicular bisector of AE, we have

$$\angle O'OO_1 = 90^\circ - \angle O_1O_2O = 90^\circ - \angle APE = 90^\circ - \angle ADQ.$$

As $OO_1 \perp AD$, this implies $\angle(O'O, OO_1) = \angle(DQ, OO_1)$. Thus, $OO' \mathbin{/\!/} DQ$ as desired. $\square$

Problem 51 (European Mathematical Cup 2019 Junior Problem 2)**.** Define a sequence (x_n) by $x_1 = \sqrt{2}$ and

$$x_{n+1} = x_n + \frac{1}{x_n}$$

for all $n \geqslant 1$. Prove that

$$\sum_{j=1}^{2019} \frac{x_j^2}{2x_j x_{j+1} - 1} > \frac{2019^2}{x_{2019}^2 + x_{2019}^{-2}}.$$

Solution. For any $n \geqslant 1$, we have $x_{n+1}^2 = \left(x_n + \frac{1}{x_n}\right)^2 = x_n^2 + \frac{1}{x_n^2} + 2$. So the right-hand side of the desired inequality is

$$\frac{2019^2}{x_{2020}^2 - 2}.$$

Also, we have

$$\frac{1}{x_{n+1}^2 - x_n^2} = \frac{1}{\frac{1}{x_n^2} + 2} = \frac{x_n^2}{1 + 2x_n^2} = \frac{x_n^2}{2x_n x_{n+1} - 1}.$$

Thus, the left-hand side of the desired inequality is

$$\sum_{j=1}^{2019} \frac{1}{x_{j+1}^2 - x_j^2}.$$

By the AM-HM inequality, we have

$$\sum_{j=1}^{2019} \frac{1}{x_{j+1}^2 - x_j^2} \geqslant \frac{2019^2}{(x_2^2 - x_1^2) + (x_3^2 - x_2^2) + \cdots + (x_{2020}^2 - x_{2019}^2)} = \frac{2019^2}{x_{2020}^2 - 2}.$$

It remains to show that equality cannot hold. Indeed, equality holds iff $x_{j+1}^2 - x_j^2$ is a constant. However, we check that $x_2 = \frac{3}{\sqrt{2}}$ and $x_3 = \frac{11\sqrt{2}}{6}$, and hence $x_3^2 - x_2^2 \neq x_2^2 - x_1^2$. This shows the inequality must be strict. $\square$

Problem 52 (Balkan MO Shortlist 2017 N5)**.** Let n be a positive integer such that $n \equiv 2 \pmod 4$. For each prime number p, let a_p be the arithmetic mean of the numbers $\left\{\frac{1^n}{p}\right\}, \left\{\frac{2^n}{p}\right\}, \ldots, \left\{\frac{(\frac{p-1}{2})^n}{p}\right\}$. Prove that there are infinitely many primes p with the same value of a_p.

Solution. Let $n = 2m$ where m is odd. Let p be any prime satisfying $p \equiv 1 \pmod 4$ and $p \equiv 2 \pmod m$. There are infinitely many such primes by the Chinese remainder theorem and Dirichlet's theorem. We claim that $a_p = \frac{1}{2}$ for all such primes p.

For any distinct $x, y \in \left\{1, 2, \ldots, \frac{p-1}{2}\right\}$, if $x^n \equiv y^n \pmod p$, then $x^{2m} \equiv y^{2m} \pmod p$. Since $x^{p-1} \equiv y^{p-1} \pmod p$ by the Fermat little theorem, and $(m, p-1) = (m, 2-1) = 1$, this implies $x^2 \equiv y^2 \pmod p$. In view of the possible values of x and y, we must have $x = y$. In other words, $1^n, 2^n, \ldots, \left(\frac{p-1}{2}\right)^n$ are pairwise distinct modulo p. Since they are perfect squares, they are exactly all possible quadratic residues modulo p. Since $p \equiv 1 \pmod 4$, every r is a quadratic residue modulo p iff $-r$ is a quadratic residue modulo p. Therefore, we can pair up these quadratic residues such that their sum is 0 modulo p. As

$$\left\{\frac{r}{p}\right\} + \left\{\frac{-r}{p}\right\} = \frac{r}{p} + \left(1 - \frac{r}{p}\right) = 1$$

if $1 \leqslant r \leqslant p-1$, the average of these two terms is $\frac{1}{2}$. It follows that $a_p = \frac{1}{2}$ as desired. $\square$